3 Working with Razor Pages

This chapter covers

- Generating HTML with Razor templates
- Learning the Razor syntax
- Working with layouts, partials, and tag helpers
- Understanding the `PageModel` as a controller and view model
- Working with handler methods and `IActionResult`

At this point, you should have a good understanding of the working parts of a Razor Pages application—how to create one, the role of the generated files, and how to configure the application's behavior through the request pipeline. Now you are ready to take a deep dive into working with the principal actor within a Razor Pages application: the Razor page itself.

As you work through the chapter, you will learn how to use Razor syntax to generate dynamic HTML and coordinate layouts and partial files to reduce your repetition of code and improve reuse. You have already had a brief introduction to layouts and partials, but to remind you, layouts act as a kind of master template for multiple pages, and partial files consist of snippets of UI that can be plugged into a host page or layout.

You will also learn about the `PageModel` class—a fundamental feature of Razor Pages that acts as both an MVC controller and a *view model*, or a container for data specific to a particular view or page. You will explore how to use the view model aspect of the `PageModel` to expose data to the Razor page in a strongly typed manner, which will improve your efficiency as a developer. Doing this will also enable you to work effectively with *tag helpers*, or components that enable server-side code to participate in the HTML generation process. You will learn about